

## 1. Confidence on a common scale

## 1. Score, prediction, and normalized margin

For binary  $h_{t(x)} \in \{-1, +1\}$ , AdaBoost's real score and prediction are

$$F_{T(x)} = \sum_{t=1}^T \alpha_t h_{t(x)}, \quad \text{sign}(F_{T(x)}).$$

For a labeled example  $(x, y)$  and nonzero coefficient vector  $\alpha$ , the **normalized functional margin** is

$$\rho(x, y) = \frac{y F_{T(x)}}{\|\alpha\|_1}, \quad \|\alpha\|_1 = \sum_{t=1}^T |\alpha_t|.$$

Its sign records correctness and its magnitude records signed-score confidence after removing arbitrary positive coefficient scale. Under  $\alpha \rightarrow c\alpha$  with  $c > 0$ , the raw score scales by  $c$ , while  $\rho$  and the prediction do not change.

## 2. Do not confuse margin with distance

The weak-hypothesis-output vector is  $h(x) = (h_1(x), \dots, h_T(x)) \in \mathbb{R}^T$ . Its unsigned  $L^\infty$  distance to the  $\alpha$ -normal hyperplane is

$$d_{\infty(h(x), \{z: \alpha \cdot z = 0\})} = \frac{|\alpha \cdot h(x)|}{\|\alpha\|_1} = \frac{|F_{T(x)}|}{\|\alpha\|_1}.$$

This distance is measured in weak-hypothesis-output space, not in the original input space and not between coefficient vectors. On a correctly classified example it equals  $\rho(x, y)$ ; on a misclassified example the distance stays positive while  $\rho(x, y) < 0$ .

## 3. When normalization gives a convex combination

If every  $\alpha_t \geq 0$  and  $\sum_t \alpha_t > 0$ , then

$$\bar{F}(x) = \frac{F_{T(x)}}{\|\alpha\|_1} = \sum_{t=1}^T \frac{\alpha_t}{\|\alpha\|_1} h_{t(x)}$$

is a convex combination of base hypotheses, and division by the positive norm preserves the sign. If signed coefficients are allowed, this convex-hull claim needs sign closure of the base class or an explicit reparameterization. The  $L^1$  denominator alone does not make negative coefficients convex weights.

## 2. Margin distributions and population control

## 1. Empirical cumulative low-margin fraction

For an i.i.d. sample  $S = ((x_i, y_i))_{i=1}^m$  and threshold  $\theta$ ,

$$\hat{R}_{S, \theta}(\bar{F}) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}[\rho(x_i, y_i) \leq \theta].$$

As  $\theta$  increases, this empirical margin distribution is nondecreasing:  $\theta_1 \leq \theta_2$  implies  $\hat{R}_{S, \theta_1} \leq \hat{R}_{S, \theta_2}$ . At zero, apart from the chosen zero-score tie convention, it recovers empirical sign error. Positive thresholds additionally count correctly classified examples with low normalized confidence.

A right-shifted training-margin curve is training evidence only. It is not a test-error curve and does not force future test error to fall.

## 2. Normalized convex ensemble class

For a binary base class  $\mathcal{H}$ ,

$$\text{conv}(\mathcal{H}) = \left\{ \sum_{j=1}^p \mu_j h_j : p \geq 1, \mu_j \geq 0, h_j \in \mathcal{H}, \sum_{j=1}^p \mu_j \leq 1 \right\}.$$

An arbitrary real linear combination is not automatically in this class. A positive-coefficient AdaBoost score enters after division by its positive  $L^1$  coefficient norm. Taking the convex hull does not increase empirical Rademacher complexity: a linear objective over convex weights is maximized at an extreme point. This is a normalized-class fact, not a monotone-rounds claim.

## 3. Fixed-threshold boosting margin bound

Let  $\mathcal{H} : \mathcal{X} \rightarrow \{-1, +1\}$  have  $d = \text{VCdim}(\mathcal{H})$  with  $1 \leq d \leq m$ . Fix  $\rho_0 > 0$  and  $\delta \in (0, 1)$ . For an i.i.d. sample  $S$  of size  $m$ , with probability at least  $1 - \delta$ , simultaneously for every  $\bar{F} \in \text{conv}(\mathcal{H})$ ,

$$R(\bar{F}) \leq \hat{R}_{S, \rho_0}(\bar{F}) + \frac{2}{\rho_0} \sqrt{\frac{2d \log(e \frac{m}{d})}{m}} + \sqrt{\frac{\log(\frac{1}{\delta})}{2m}},$$

where  $R(\bar{F}) = \Pr[y\bar{F}(x) \leq 0]$ . The right side contains the empirical low-margin mass, a base-class complexity term, and a confidence term. It is an upper bound, not observed population error or an equality.

## 4. Read the quantifiers and tradeoff correctly

The order is: fix  $\rho_0 > 0$ ; draw the i.i.d. sample; then, with probability at least  $1 - \delta$ , assert the inequality uniformly over normalized ensemble functions. The displayed event is not uniform over every threshold. Choosing a favorable threshold after seeing the sample needs the corrected all-threshold theorem with its additional log-log term. Increasing  $\rho_0$  can increase the cumulative empirical term but decreases the  $\frac{1}{\rho_0}$  complexity penalty. Increasing  $m$  shrinks both nonempirical square-root terms on an approximate  $\frac{1}{\sqrt{m}}$  scale, with logarithmic dependence retained. The displayed bound has no explicit round count  $T$ , but does not imply that more rounds improve margins or test error.

## 3. Exponential loss as the training surrogate

## 1. Pointwise and empirical upper bounds

For signed score  $z = yF(x)$ , pointwise

$$\mathbf{1}[z \leq 0] \leq e^{-z}.$$

If  $z \leq 0$ , both sides are at least one; if  $z > 0$ , the indicator is zero. Averaging gives  $\hat{R}_{S(\text{sign}(F))} \leq m^{-1} \sum_i e^{-y_i F(x_i)}$ . Exponential loss is smooth and convex, but those facts, domination, calibration, robustness, and population performance are distinct. This empirical inequality does not certify test error.

## 4. One AdaBoost round as one coordinate step

## 1. Objective and current example distribution

Fix a finite binary dictionary  $\mathcal{H} = \{h_1, \dots, h_N\}$ . With  $\beta \in \mathbb{R}^N$ ,

$$F_{\beta}(x) = \sum_{j=1}^N \beta_j h_{j(x)}, \quad \mathcal{L}(\beta) = \frac{1}{m} \sum_{i=1}^m e^{-y_i F_{\beta}(x_i)}.$$

At round  $t$ , normalize the current exponential masses:

$$D_{t(i)} = \frac{e^{-y_i F_{t-1}(x_i)}}{\sum_{r=1}^m e^{-y_r F_{t-1}(x_r)}}.$$

Coordinate descent changes one coefficient while holding all others fixed. The variant used here selects a maximum-descent coordinate in this fixed dictionary, then performs exact line minimization.

## 2. Weighted error selects the coordinate

For coordinate  $k$ , define

$$\varepsilon_{t,k} = \sum_i D_{t(i)} \mathbf{1}[h_{k(x_i)} \neq y_i].$$

Let  $C_t = \mathcal{L}(\beta^{t-1}) > 0$ . Moving along  $h_k$  gives

$$\frac{d}{d\eta} \mathcal{L}(F_{t-1} + \eta h_k)|_{\eta=0} = C_t(2\varepsilon_{t,k} - 1).$$

The factor  $C_t$  is positive and common to all coordinates, so minimum current weighted error is exactly maximum immediate descent. It may be ignored for coordinate ranking, but is not generally equal to one and must remain in the derivative equality.

## 3. Exact line minimum and round factor

For a selected coordinate with weighted error  $\varepsilon_t$ , the line objective is

$$\mathcal{L}(F_{t-1} + \eta h_t) = C_t((1-\varepsilon_t)e^{-\eta} + \varepsilon_t e^{\eta}).$$

For  $0 < \varepsilon_t < \frac{1}{2}$ , strict convexity gives the unique finite minimum

$$\alpha_t = \frac{1}{2} \log \frac{1-\varepsilon_t}{\varepsilon_t} > 0.$$

The minimum factor is  $2\sqrt{\varepsilon_t(1-\varepsilon_t)}$ . Exact exponential-loss line search, not convexity alone, licenses this coefficient.

## 4. Conditional equivalence

One greedy coordinate step coincides with one AdaBoost round only when:

1. base hypotheses come from the fixed finite binary dictionary (or a separately defined oracle analogue);
2. the coordinate has minimum current  $D_t$ -weighted error;
3.  $0 < \varepsilon_t < \frac{1}{2}$ ; and
4. line search on the empirical exponential objective is exact.

Approximate weak-learner selection or line search gives only an approximate greedy step. Cyclic, random, or block-coordinate descent, and another optimizer of the same loss, need not reproduce AdaBoost's path, coefficients, classifier, margins, or generalization behavior.

## 5. Boundary regimes

| Error                             | Best step                  | Meaning                                                             |
|-----------------------------------|----------------------------|---------------------------------------------------------------------|
| $\varepsilon_t = 0$               | $\eta \rightarrow +\infty$ | no finite log-odds minimizer                                        |
| $\varepsilon_t = \frac{1}{2}$     | $\eta = 0$                 | zero step; no positive progress                                     |
| $\frac{1}{2} < \varepsilon_t < 1$ | $\eta < 0$                 | negative direction; use sign closure or explicit reparameterization |

Do not silently substitute any boundary row into the finite positive log-odds formula.

## 5. Persistent hard examples and evidence

## 1. What a large training weight means

Because  $D_{t(i)}$  is proportional to  $e^{-y_i F_{t-1}(x_i)}$ , a persistently negative or small signed score can concentrate later weight and influence weak-hypothesis selection. This shows current difficulty, not its cause: the same pattern can arise from a wrong label, a rare but valid region, an insufficient weak class, or early optimization.

AdaBoost is conditionally sensitive to persistent hard examples, especially when the weak class cannot resolve them. A large weight neither identifies label noise nor establishes a universal noise threshold or monotone test-error worsening.

## 2. Match each claim to its evidence

| Claim       | Inspect             | Evidence                                     |
|-------------|---------------------|----------------------------------------------|
| round       | example             | $D_t$ update and normalized                  |
| mechanics   | reweighting         | weight trace                                 |
| objective   | optimization        | loss, direction, and exact line calculation  |
| decrease    |                     |                                              |
| population  | margin              | fixed-threshold theorem with every condition |
| certificate |                     |                                              |
| fitted test | external validation | untouched held-out or                        |
| behavior    |                     | deployment data                              |

Equal empirical exponential losses do not imply equal coefficients, classifiers, margins, or test errors. A training objective, a population bound, and observed validation answer different questions; none automatically supplies the others' conclusion.